Analytical Solving — Activity

Given:-

$G_x = 24$

$G_y = 18$

Gradient magnitude :

$$|G| = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{24^2 + 18^2}$$

$$= \sqrt{900}$$

$$= 30$$

$$|G| = 30$$

$$\theta = \tan^{-1}\left(\frac{G_x}{G_y}\right)$$

$$= \tan^{-1}\left(\frac{24}{18}\right)$$

$$= 53.130^\circ \; //$$

②

Pixel	Magnitude
P_1	25
P_2	50
P_3	90
P_4	60
P_5	30

Gradient direction = Horizontal

Pixel	Magnitude	left neighbour	Righ neig hbour	Non's output
P_1	25	none	P_2 (50)	0
P_2	50	P_1 (25)	P_3 (90)	0
P_3	90	P_2 (50)	P_4 (60)	90
P_4	60	P_3 (90)	P_5 (30)	0

Double thresholding

- High threshold = 100
  low threshold = 50

strong edge $|G| \geq T\_H$

Weak edge $T\_L \leq |G| < T\_H$

Non - edge $|G| < T\_L$

Pixel	Magnitude			
P_1	135	$135 > 100$	$\rightarrow$	strong
P_2	90	$50 < 90 < 100$	$\rightarrow$	Weak
P_3	45	$45 < 50$	$\rightarrow$	None
P_4	120	$120 > 100$	$\rightarrow$	strong
P_5	70	$50 < 70 < 100$	$\rightarrow$	Weak
P_6	20	$20 < 50$	$\rightarrow$	None.

④ Given:

$T\_H = 100$

$T\_L = 50$

Pixel	Magnitude
P_1	130
P_2	95
P_3	40
P_4	115
P_5	65
P_6	25
P_7	105

P_2 is Connected to P_1

P_8 is not Connected to any strong edge.

Strong edges

$P_1 - 130 > 100 \rightarrow$ strong

$P_4 - 115 > 100 \rightarrow$ strong

$P_7 - 105 > 100 \rightarrow$ strong

Weak edge :-

$P_2 - 50 < 95 < 100 \rightarrow$ Weak

$P_5 - 50 < 65 < 100 \rightarrow$ Weak

Non-edge

$P_3 - 40 < 50$

$P_6 - 25 < 50$

Final edge pixels after Hysteresis

$P_1 \rightarrow$ Keep

$P_2 \rightarrow$ Connected to (P_1)

$P_4 \rightarrow$ Keep

$P_5 \rightarrow$ removed

$P_7 \rightarrow$ Keep.